

Adaptive Hue-Preserving Contrast Enhancement

with Automatic Parameter Selection



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Outline

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- 3 Adaptive σ Selection
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- 5 Interactive Gradio Tool
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The Challenge

Manual Parameter Tuning

σ and λ must be hand-picked per image

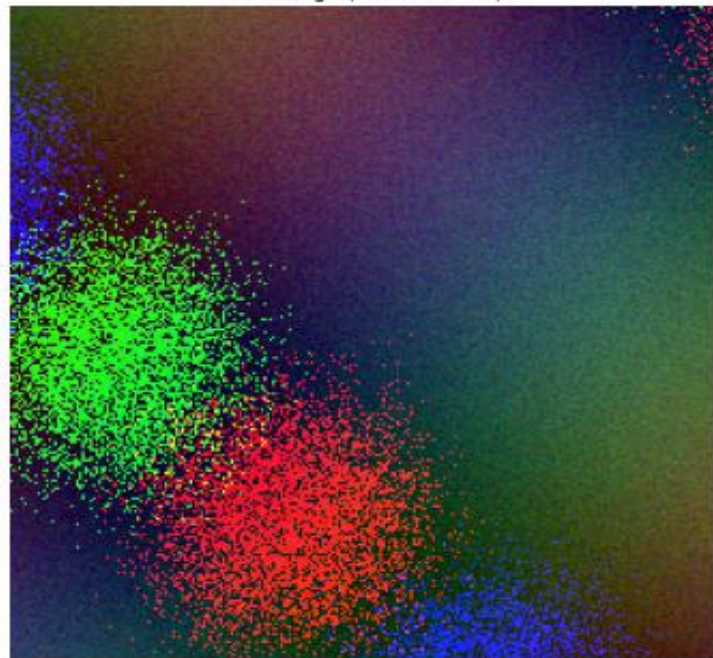
Gamut Violation

RGB values can exceed valid $[0, 255]$ range

Hue Distortion

Aggressive enhancement shifts perceived colors

Test Image (Low Contrast)



Matsumoto et al. (ICIP 2024)

1

Histogram Specification

Gaussian-smoothed target histogram per RGB channel



2

Hue-Preserving Transformation

Linear model $O = A I + B e$ with λ chroma control



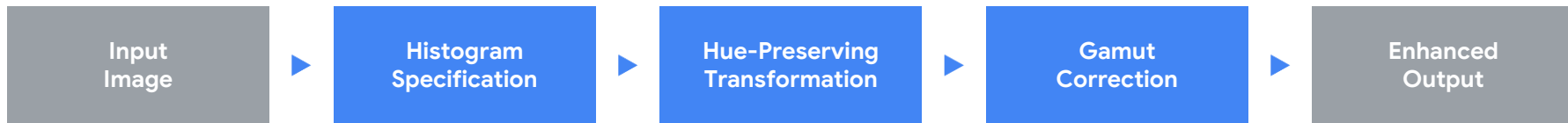
3

Gamut Correction

Equi-hue plane projection for out-of-range pixels



Algorithm Pipeline



Processing Pipeline ($\lambda=0$, neutral)

Step 0: Original Image



Step 1: Histogram Specification



Step 2: Hue Preservation (pre-correction)



Step 3: Gamut Correction (final)

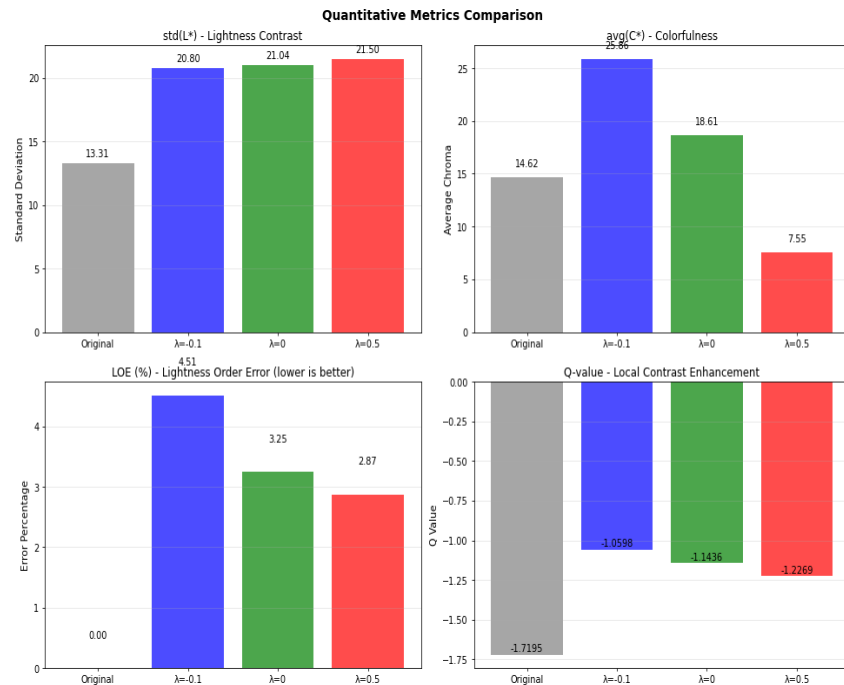


- σ controls smoothing
- λ controls chroma

Novel: Adaptive σ Selection

- Compute Shannon entropy of each RGB histogram
- Low entropy (narrow histogram) $\rightarrow$ needs more smoothing
- High entropy (spread histogram) $\rightarrow$ needs less smoothing
- Map average entropy [3, 8] $\rightarrow \sigma$ [25, 75]

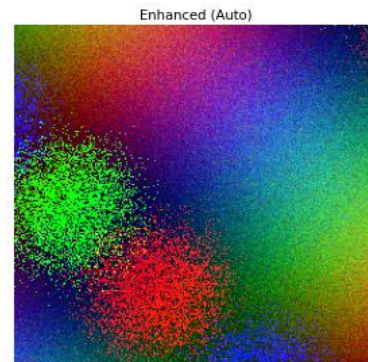
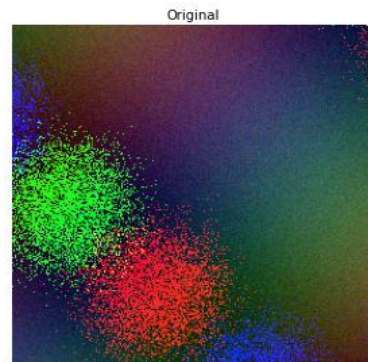
$$H = -\sum p(x) \log_2 p(x) \rightarrow \sigma = 25 + (H - 3) \times 10$$



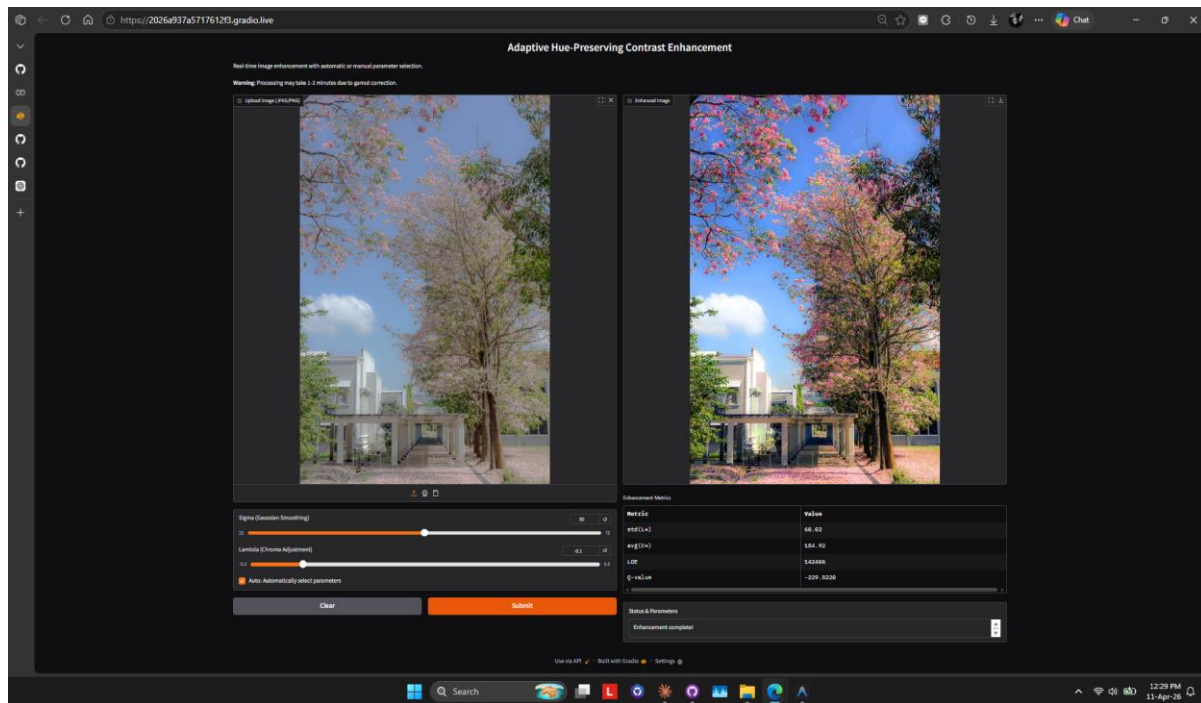
Novel: Adaptive λ Selection

- Compute Hasler-Süsstrunk colorfulness metric
- Low colorfulness (< 15) $\rightarrow \lambda = -0.1$ (boost saturation)
- Medium (15–40) $\rightarrow \lambda \approx 0$ (neutral)
- High colorfulness (> 40) $\rightarrow \lambda$ up to $+0.3$ (reduce)

$$CF = \sqrt{(\sigma^2_{rg} + \sigma^2_{yb})} + 0.3 \times \sqrt{(\mu^2_{rg} + \mu^2_{yb})}$$

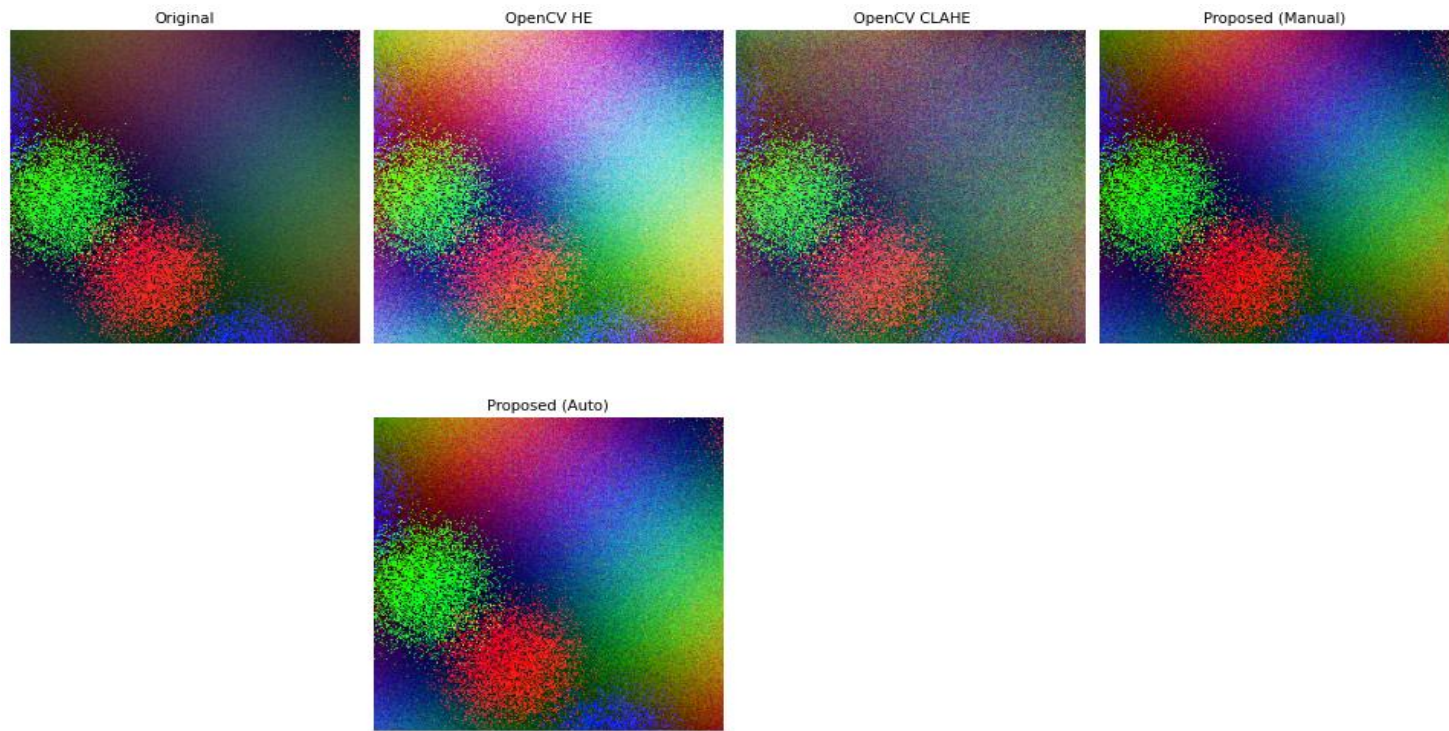


Gradio Interactive Interface



Real-time parameter exploration via Google Colab

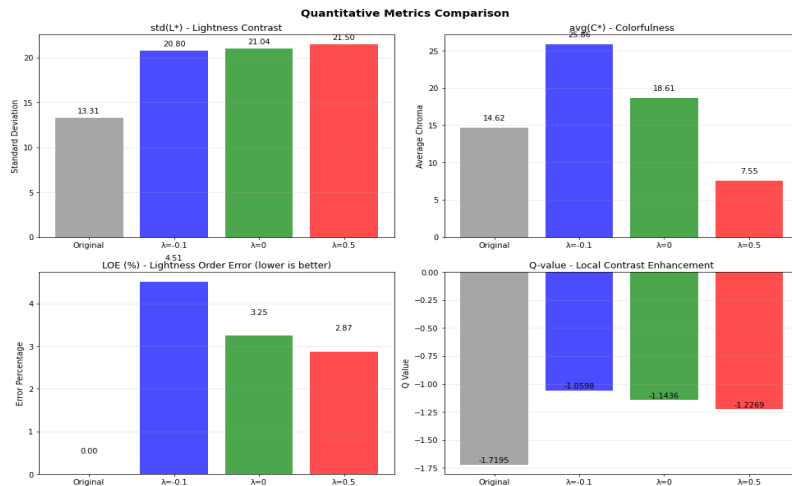
Visual Comparison



Enhancement comparison across methods on two test images

Quantitative Evaluation

Method	std(L*)	avg(C*)	LOE ↓	Q ↑
OpenCV HE	53.13	196.83	764240	-206.15
OpenCV CLAHE	41.17	189.97	854505	-164.87
Manual ($\sigma=50, \lambda=-0.1$)	53.01	197.16	420337	-17.10
Auto (ours)	54.33	198.34	490140	-24.63



↑ **Q-value** Higher is better
↓ **LOE** Lower is better

Key Findings



Auto parameters achieve comparable or better results than manual tuning



Hue preservation is maintained throughout the entire pipeline



Interactive Gradio tool enables real-time exploration on any device



Method generalises across portraits, landscapes, and low-light scenes

Conclusion & Future Work

Achievements

- Automated σ and λ selection — no manual tuning
- Hue preservation maintains color fidelity
- Practical interactive tool for real-world use
- Robust across diverse image types

Future Directions

- Deep learning-based parameter prediction
- Extension to video with temporal consistency
- Mobile app for real-time enhancement
- Integration with image editing software

Thank You — Questions?